

MML Chapter 2

Detailed worked answers · explanation, elimination steps, then the result

2.1 · Custom operation $a * b = ab + a + b$ on $\mathbb{R} \setminus \{-1\}$

TOPICS & DEFINITIONS

Group — a set with an operation obeying 4 axioms: closure, associativity, identity, inverses

Abelian group — a group whose operation is also commutative ($a * b = b * a$)

Closure — combining two elements always lands back inside the set

Identity e — the element that leaves others unchanged: $a * e = a$

Inverse — the element that undoes a , returning the identity

The key move is factoring: $a * b = (a+1)(b+1) - 1$. So the operation behaves like ordinary multiplication shifted by 1, which makes every axiom easy.

Closed (never gives -1), associative, identity 0, and every element has an inverse. Commutative too, so Abelian. Part b: $3 * x * x = 4(x+1)^2 - 1 = 15$.

ANSWER

$(\mathbb{R} \setminus \{-1\}, *)$ is an Abelian group. Identity $e = 0$, inverse $a^{-1} = -a/(a+1)$.

2.1b: $x = 1$ or $x = -3$

2.2 · Integers mod n under $\oplus$

TOPICS & DEFINITIONS

Congruence class $[a]$ — all integers leaving the same remainder as a when divided by n

Modular addition $\oplus$ — $[a] \oplus [b] = [a+b]$, then reduce mod n

Inheritance — Z_n is a group because ordinary integer addition already is

Prime condition — $(Z_n \setminus \{0\}, \otimes)$ is a group only when n is prime (every element then has a multiplicative inverse)

$Z_n = \{[0], [1], \dots, [n-1]\}$ with $[a] \oplus [b] = [a+b]$, reduced mod n . It's a group because ordinary integer addition is, and the classes inherit that.

Identity $[0]$, inverse of $[a]$ is $[n-a]$, commutative. The multiplicative version $(Z_n \setminus \{0\}, \otimes)$ is a group only when n is prime.

ANSWER

$(Z_n, \oplus)$ is an Abelian group. Identity $[0]$, inverse $[n-a]$.

$(Z_n \setminus \{0\}, \otimes)$ is a group iff n is prime.

2.3 · Upper-unitriangular 3×3 matrices under multiplication

TOPICS & DEFINITIONS

Matrix group — a set of matrices closed under multiplication that satisfies the 4 group axioms

Upper unitriangular — 1s on the diagonal, 0s below, free entries above

Associativity — matrix multiplication is always associative (free axiom)

Non-commutative — for matrices $AB \neq BA$ in general, so a group can be non-Abelian

G is all matrices with 1s on the diagonal, 0s below, free x, y, z above. Multiplying two keeps the shape (closed), multiplication is associative, the identity is in G , and inverses keep the shape too.

So it's a group, but not Abelian: the (1,3) entry of a product depends on the order, so a concrete pair multiplied both ways differs.

ANSWER

$(G, \cdot)$ is a group, not Abelian.

2.4 · Matrix products (check dimensions first!)

TOPICS & DEFINITIONS

Dimension rule — AB is defined only if A 's columns = B 's rows: $(m \times n)(n \times p) \rightarrow (m \times p)$

Entry rule — entry $(i,j) = \text{row } i \text{ of } A \cdot \text{column } j \text{ of } B$ (multiply pairwise, sum)

Non-commutative — $AB \neq BA$ in general (compare parts b and c: same matrices, swapped, different results)

A product AB is only defined when A 's column count equals B 's row count. Part a fails this test, so it has no answer; the rest multiply normally (row of A dotted with column of B).

a. 3×2 times 3×3 : inner dimensions 2 and 3 don't match $\rightarrow$ **undefined**.

ANSWER

a) undefined (dimension mismatch)

b) [4 3 5 ; 10 9 11 ; 16 15 17]

c) [5 7 9 ; 11 13 15 ; 8 10 12]

d) [14 6 ; -21 2]

e) [12 3 -3 -12 ; -3 1 2 6 ; 6 5 1 0 ; 13 12 3 2]

2.5 · Find all solutions of $Ax = b$

TOPICS & DEFINITIONS

Gaussian elimination — row operations to reach echelon (staircase) form

Contradiction row — $0 = \text{nonzero}$ means the system is inconsistent $\rightarrow$ no solution

Free variable — a column with no pivot; can be anything $\rightarrow$ infinitely many solutions

Solution set S — could be empty, a single point, or an infinite family

Solved by Gaussian elimination on $[A \mid b]$. A contradiction row ($0 = \text{nonzero}$) means no solution; free variables mean infinitely many.

ANSWER

2.5a: no solution, $S = \emptyset$ (a contradiction row appears).

2.5b: infinitely many, two free parameters.

2.6 · Solving $Ax = b$ with Gaussian elimination

TOPICS & DEFINITIONS

RREF — reduced row echelon form: pivots are 1, alone in their column, stepping down-right

Pivot vs free column — pivot columns $\rightarrow$ basic variables; no-pivot columns $\rightarrow$ free variables

General solution — one particular solution + a combination of the free-variable directions

Row-reduce $[A \mid b]$ to RREF. Pivots land in columns 2, 4, 5; columns 1, 3, 6 have no pivot, so they're free variables. Reading off the rows gives each basic variable in terms of the free ones.

RREF OF $[A \mid b]$

$$\left[\begin{array}{cccccc|c} 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]$$

pivots in columns 2, 4, 5 $\rightarrow x_2, x_4, x_5$ basic; x_1, x_3, x_6 free

ANSWER

$$x = (0, 1, 0, -2, 1, 0) + x_1(1, 0, 0, 0, 0, 0) + x_3(0, 0, 1, 0, 0, 0) + x_6(0, -1, 0, -1, 1, 1), \quad x_1, x_3, x_6 \in \mathbb{R}$$

2.7 · Solving $Ax = 12x$

TOPICS & DEFINITIONS

Eigenvalue / eigenvector — $Ax = \lambda x$: a vector A only scales (never rotates); λ is the scale factor

Rearrangement — $Ax = \lambda x \rightarrow (A - \lambda I)x = 0$, a homogeneous system

Eigenspace — all eigenvectors for a given λ (a line or plane through the origin)

Constraint — an extra equation like $\sum x_i = 1$ selects one point on that line

Rearrange to $(A - 12I)x = 0$, then eliminate. A zero row appears, so x_3 is free and the solutions form a line, every multiple of $(3,3,2)$. The sum-to-1 condition fixes $t = 1/8$.

$(A - 12I)x = 0$ · ELIMINATION

$$\begin{bmatrix} -6 & 4 & 3 \\ 6 & -12 & 9 \\ 0 & 8 & -12 \end{bmatrix}$$

$\rightarrow$

$$\begin{bmatrix} -6 & 4 & 3 \\ 0 & -8 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

zero row $\rightarrow x_3$ free. $x_2 = (3/2)x_3$, $x_1 = (3/2)x_3 \rightarrow x = t(3,3,2)$

constraint $3t + 3t + 2t = 1 \rightarrow t = 1/8$

ANSWER

$x = (3/8, 3/8, 1/4)$

2.8 · Inverses

TOPICS & DEFINITIONS

Inverse A^{-1} — the matrix with $A \cdot A^{-1} = I$ (the identity)

Determinant — a single number measuring how much A scales space

Invertible iff $\det \neq 0$ — $\det = 0$ means A squashes a dimension $\rightarrow$ no inverse (singular)

Gauss-Jordan method — reduce $[A \mid I]$ until the left is I ; the right becomes A^{-1}

a. $\det = 0$ (row1 + row3 = 2×row2), so no inverse.

b. Augment $[A \mid I]$ and reduce the left to the identity; the right becomes A^{-1} . $\det = 1$, so it's invertible.

$[A \mid I] \rightarrow [I \mid A^{-1}]$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & -2 \end{array} \right]$$

ANSWER

a) not invertible ($\det = 0$).

b) $A^{-1} = [0 \ -1 \ 0 \ 1; -1 \ 0 \ 0 \ 1; 1 \ 1 \ 0 \ -1; 1 \ 1 \ 1 \ -2]$

2.9 · Which sets are subspaces of $\mathbb{R}^3$?

TOPICS & DEFINITIONS

Subspace — a subset of $\mathbb{R}^3$ containing 0 and closed under addition and scaling

The 3 tests — (1) zero vector in it, (2) closed under +, (3) closed under scalar $\times$

Span — all linear combinations of given vectors — always a subspace

The 3 killers — a square term, an integer-only condition, or $=$ a nonzero constant

Subspace = contains 0, closed under addition and scaling. Breakers: a square, an integer-only rule, or $=$ a non-zero constant.

a. span of two vectors $\rightarrow$ yes. **b.** λ^2 can't be negative, scaling by -1 escapes $\rightarrow$ no. **c.** plane; through origin only if $\gamma = 0$. **d.** integer middle coord; scale by 0.5 escapes $\rightarrow$ no.

ANSWER

a) subspace · b) not · c) subspace iff $\gamma = 0$ · d) not

2.10 · Linear independence

TOPICS & DEFINITIONS

Linearly independent — the only combination giving 0 is all-zero coefficients

Linearly dependent — some vector is a combination of the others (redundant)

Method — vectors as columns, row-reduce, count pivots

Rule — pivot in every column → independent; a missing column pivot → dependent

Vectors as columns, row-reduce, count pivots. Pivot in every column = independent; a missing column pivot = dependent (zero rows alone don't count).

a) vectors as columns → reduce

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 1 & -3 \\ 3 & -2 & 8 \end{bmatrix}$$

→

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

a) only 2 pivots, a zero row → **dependent**. b) 3 pivots (rank 3) → **independent**

ANSWER

a) dependent ($x_3 = 2x_1 - x_2$) b) independent (rank 3)

2.11 · Write y as a linear combination

TOPICS & DEFINITIONS

Linear combination — $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3$: scale each vector, then add

Set-up — build $[v_1 \ v_2 \ v_3 \mid y]$ — same as solving a linear system

Method — reduce to RREF; the coefficients appear in the final column

Uniqueness — a unique answer exists because the vectors are independent ($\det \neq 0$)

Set up $[x_1 \ x_2 \ x_3 \mid y]$ and reduce to RREF; the coefficients appear in the last column. Check:
 $-6(1)+3(1)+2(2) = 1$ matches y .

$[x_1 \ x_2 \ x_3 \mid y] \rightarrow \text{RREF}$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & -1 & -2 \\ 1 & 3 & 1 & 5 \end{bmatrix}$$

$\rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

coefficients read straight off the last column

ANSWER

$$y = -6x_1 + 3x_2 + 2x_3$$